

Literature-Implied Negative Answer for arXiv:1612.05784: Flat Tight Frames and Partial Hadamard Completion

Run fa_banach_001, agent_lane_05

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Status and Source Conjecture

This is a `literature_implied_answer` (full negative answer) packet. After Theorem 4.4, on page 17 of the locally rendered source PDF, King conjectures that every real flat tight frame comes from an appropriately scaled submatrix of a Hadamard matrix [1]. The preceding sentence makes the intended construction explicit: remove rows from a square Hadamard matrix and scale the remaining matrix so that its columns have norm one.

Identification with Partial Hadamard Completion

Let L be the $k \times N$ synthesis matrix of a real flat unit-norm tight frame. Flatness and unit column norms give

$$L = \frac{1}{\sqrt{k}}H, \quad H \in \{\pm 1\}^{k \times N}.$$

Tightness is equivalent to

$$HH^\top = NI_k.$$

Thus H is a row-Hadamard, or partial Hadamard, matrix. It is obtained by deleting rows from a Hadamard matrix of order N exactly when it extends to such a square Hadamard matrix (up to row and column signs and permutations).

Bulutoglu and Kaziska constructed a maximal 13×32 row-Hadamard matrix [2]. Consequently, scaling their matrix by $1/\sqrt{13}$ gives a real flat unit-norm tight frame of 32 vectors in $\mathbb{R}^{13}$ with frame operator $(32/13)I_{13}$. Since the sign matrix is maximal—not even one orthogonal sign row can be appended—it cannot be a row-deleted submatrix of an order-32 Hadamard matrix. This answers the source conjecture negatively. The relation is agent-identified rather than claimed in the supporting paper, so the provenance is `literature_implied_answer`.

A Four-Row Exact Certificate

There is also a short explicit witness. With

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad H = H_4 \otimes \begin{pmatrix} 1 & 1 & 1 \end{pmatrix},$$

we have $HH^\top = 12I_4$. Hence $L = H/2$ is a real flat unit-norm tight frame of twelve vectors in $\mathbb{R}^4$, with frame operator $3I_4$.

To see that H cannot be extended by even one sign row, suppose $x \in \{\pm 1\}^{12}$ is orthogonal to every row. Split x into four consecutive three-entry blocks and let s_j be the sum of block j . Orthogonality is precisely $H_4 s = 0$. Since H_4 is invertible, $s = 0$. But every s_j is a sum of three signs and is therefore odd, a contradiction. Thus H is maximal and cannot arise by row deletion from an order-12 Hadamard matrix. The included verifier checks the Gram identity and all 2^{12} possible extension rows.

This certificate is not presented as novel. The 2009 publisher record notes that Beder had already constructed maximal four-row examples for every odd multiple of four columns. It is included only to make the negative answer self-contained and transparent.

Search Bounds and Scope

Cheap run indexes were searched for the arXiv id, title, flat tight frames, partial Hadamard matrices, and Hadamard completion; no duplicate packet was found. A bounded primary-literature search for the exact conjecture and for nonextendible partial Hadamard matrices found the 2009 result above and earlier partial-Hadamard literature. The decisive paper is paywalled, so no local supporting PDF was available; its DOI publisher record gives the abstract, metadata, and maximal 13×32 result. This packet addresses the literal flat-tight-frame conjecture, not classifications under extra hypotheses such as distinct projective frame vectors or equiangularity.

References

- [1] E. J. King, *New Constructions and Characterizations of Flat and Almost Flat Grassmannian Fusion Frames*, arXiv:1612.05784.
- [2] D. A. Bulutoglu and D. M. Kaziska, *A counterexample to Beder's conjectures about Hadamard matrices*, J. Statist. Plann. Inference **139** (2009), 3381–3383, doi:10.1016/j.jspi.2009.02.003.